CPUE project 2

Beth Babcock

2025-11-20

```
library(tidyverse)
library(glmmTMB)
library(MuMIn)
library(DHARMa)
theme_set(theme_bw())
```

1.Use the same SWO data we used in the last assignment. Fit a Tweedie model to the CPUE (SWO/set) and used dredge to find the best set of predictor variables. Look at the DHARMa residuals. Does it fit well?

```
obsdat<-read_csv("obsdatSwo.csv")%>%
  mutate(Year=factor(Year),
         season=factor(season),
         area=factor(area),
         cpue=SWO/sets)
summary(obsdat)
```

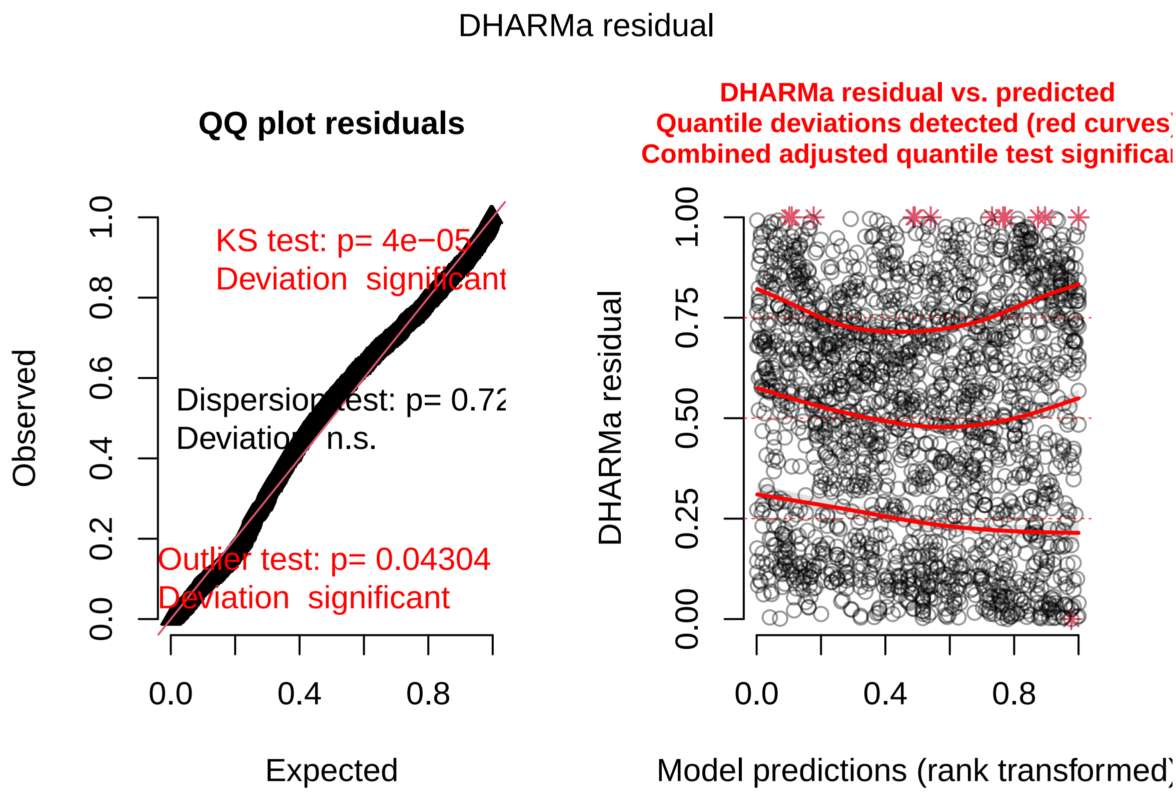
```
##       Year       area     season        lat                lon
##  1994   : 104   N: 983   1:605   Min.   :-47.500   Min.   :-81.50
##  1996   : 102   S:1000   2:475   1st Qu.:-11.500   1st Qu.:-27.50
##  1990   :  89            3:384   Median : -0.500   Median :-17.50
##  1991   :  89            4:519   Mean   :  1.998   Mean   :-16.19
##  1995   :  86                    3rd Qu.: 14.500   3rd Qu.:  2.00
##  2004   :  85                    Max.   : 49.500   Max.   : 19.50
##  (Other):1428
##       hbf             BUM              SWO              sets          fleet
##  Min.   : 3.00   Min.   :  0.000   Min.   :  0.00   Min.   : 1.00   Min.   :2
##  1st Qu.:10.00   1st Qu.:  0.000   1st Qu.:  7.00   1st Qu.:11.00   1st Qu.:2
##  Median :10.00   Median :  2.000   Median : 21.00   Median :20.00   Median :2
##  Mean   :14.98   Mean   :  6.933   Mean   : 29.88   Mean   :19.88   Mean   :2
##  3rd Qu.:24.00   3rd Qu.:  8.000   3rd Qu.: 38.00   3rd Qu.:20.00   3rd Qu.:2
##  Max.   :24.00   Max.   :140.000   Max.   :364.00   Max.   :99.00   Max.   :2
##
##       cpue
##  Min.   :0.0000
##  1st Qu.:0.6962
##  Median :1.3333
##  Mean   :1.4959
##  3rd Qu.:2.0000
##  Max.   :9.0000
##
```

```r
Tweedie1<-glmmTMB(cpue~Year+area+season,
                  data=obsdat,
                  family="tweedie",
                  na.action="na.fail")
TweedieDredge<-dredge(Tweedie1,
                      rank="BIC")
head(TweedieDredge)
```

```
## Global model call: glmmTMB(formula = cpue ~ Year + area + season, data = obsdat,
##     family = "tweedie", na.action = "na.fail", ziformula = ~0,
##     dispformula = ~1)
## ---
## Model selection table
##   cnd((Int)) dsp((Int)) cnd(are) cnd(ssn) cnd(Yer) df    logLik    BIC  delta
## 2     0.2330          +        +                   4 -2800.449 5631.3   0.00
## 4     0.2820          +        +        +          7 -2793.545 5640.2   8.97
## 6     0.3272          +        +                 + 32 -2724.945 5692.8  61.58
## 8     0.3608          +        +        +        + 35 -2717.498 5700.7  69.46
## 1     0.4027          +                            3 -2848.691 5720.2  88.89
## 3     0.4298          +                 +          6 -2843.722 5733.0 101.73
##   weight
## 2  0.989
## 4  0.011
## 6  0.000
## 8  0.000
## 1  0.000
## 3  0.000
## Models ranked by BIC(x)
```

```r
TweedieBest<-get.models(TweedieDredge,subset=4)[[1]]  #Select best with a year effect
plot(simulateResiduals(TweedieBest))
```

```
## Warning in newton(lsp = lsp, X = G$X, y = G$y, Eb = G$Eb, UrS = G$UrS, L = G$L,
## : Fitting terminated with step failure - check results carefully
```




```r
summary(TweedieBest)
```

```
##  Family: tweedie  ( log )
## Formula:          cpue ~ area + season + Year + 1
## Data: obsdat
##
##      AIC      BIC   logLik -2*log(L)  df.resid
##   5505.0   5700.7  -2717.5    5435.0      1948
##
##
## Dispersion parameter for tweedie family (): 0.582
##
## Conditional model:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.36076    0.07380   4.888 1.02e-06 ***
## areaS        0.29014    0.03108   9.336  < 2e-16 ***
## season2     -0.02061    0.04141  -0.498 0.618725
## season3     -0.01708    0.04403  -0.388 0.698084
## season4     -0.14509    0.04095  -3.543 0.000396 ***
## Year1991     0.16433    0.09520   1.726 0.084303 .
## Year1992     0.33002    0.09982   3.306 0.000946 ***
## Year1993     0.21971    0.09639   2.279 0.022653 *
## Year1994     0.16403    0.09220   1.779 0.075250 .
## Year1995     0.19904    0.09588   2.076 0.037897 *
## Year1996    -0.17313    0.09769  -1.772 0.076350 .
```

```
## Year1997    -0.15727    0.10587  -1.485 0.137416
## Year1998    -0.18829    0.10375  -1.815 0.069542 .
## Year1999    -0.14393    0.10465  -1.375 0.169028
## Year2000    -0.09111    0.10235  -0.890 0.373337
## Year2001     0.05437    0.10950   0.497 0.619536
## Year2002    -0.00788    0.10944  -0.072 0.942598
## Year2003    -0.32911    0.10871  -3.027 0.002467 **
## Year2004    -0.17523    0.10287  -1.703 0.088490 .
## Year2005    -0.17964    0.11386  -1.578 0.114624
## Year2006    -0.20178    0.10733  -1.880 0.060105 .
## Year2007    -0.29527    0.11047  -2.673 0.007518 **
## Year2008    -0.24258    0.10646  -2.279 0.022689 *
## Year2009    -0.23807    0.11455  -2.078 0.037681 *
## Year2010    -0.36487    0.11992  -3.043 0.002346 **
## Year2011    -0.21736    0.11821  -1.839 0.065949 .
## Year2012    -0.03393    0.11317  -0.300 0.764292
## Year2013    -0.11299    0.12282  -0.920 0.357589
## Year2014    -0.35056    0.12823  -2.734 0.006261 **
## Year2015    -0.12818    0.13038  -0.983 0.325534
## Year2016    -0.15395    0.13039  -1.181 0.237720
## Year2017    -0.07973    0.12370  -0.645 0.519238
## Year2018    -0.31921    0.12797  -2.494 0.012619 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

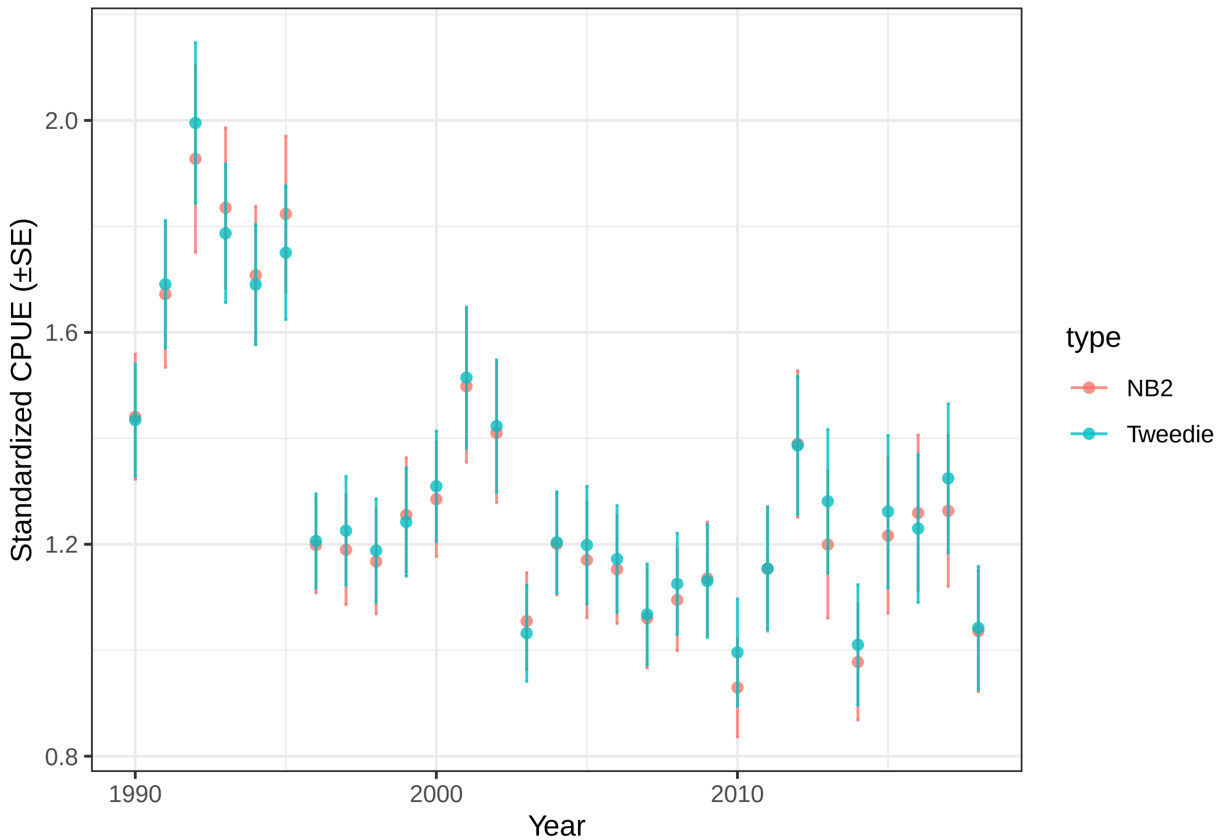
The best model did not include the year effect. Because year is the most important variable in the analysis, we can decide to use the best model with a year effect, which is the 4th one.

2. Plot the predicted mean CPUE per year plus and minus SE for the negative binomial 2 fit from the last assignment and the Tweedie you just fit. How similar are they?

```r
NB2<-glmmTMB(SWO~Year+area+season+offset(log(sets)),
                  data=obsdat,
                  family="nbinom2")
newdata1<-newdata2<-expand.grid(Year=factor(levels(obsdat$Year)),
                    area=factor("N"),
                    season=factor("1"),
                    sets=1)  #1 set to get CPUE
temp<-predict(NB2,
              newdata = newdata1,
              se.fit = TRUE,
              type="response")
newdata1$fit<-temp$fit
newdata1$se<-temp$se.fit
temp<-predict(TweedieBest,
              newdata = newdata2,
              se.fit = TRUE,
              type="response")
newdata2$fit<-temp$fit
newdata2$se<-temp$se.fit

bind_rows(list(NB2=newdata1,
          Tweedie=newdata2),.id="type") %>%
```

```r
ggplot(aes(x=as.numeric(as.character(Year)),y=fit,ymin=fit-se,
           ymax=fit+se,color=type))+
  geom_point(alpha=0.8)+
  geom_errorbar(width=0.1,alpha=0.8)+
  labs(x="Year",
    y="Standardized CPUE (±SE)")
```



Very similar results.

3. The swordfish data is not appropriate for delta-lognormal, because nearly all the trips caught at least one swordfish. For this problem, we will use a new BUM data set called BUM2.csv, which is simulated data for a different fleet only in the South Atlantic, and with possible predictor variables hbf(hooks between floats, numeric), Year and season. Fit the binomial model, look at the residuals and use dredge to find the best model. Do the same for the lognormal on the positive observations. Make the combined index and plot it plus and minus standard error. What is the trend over time?

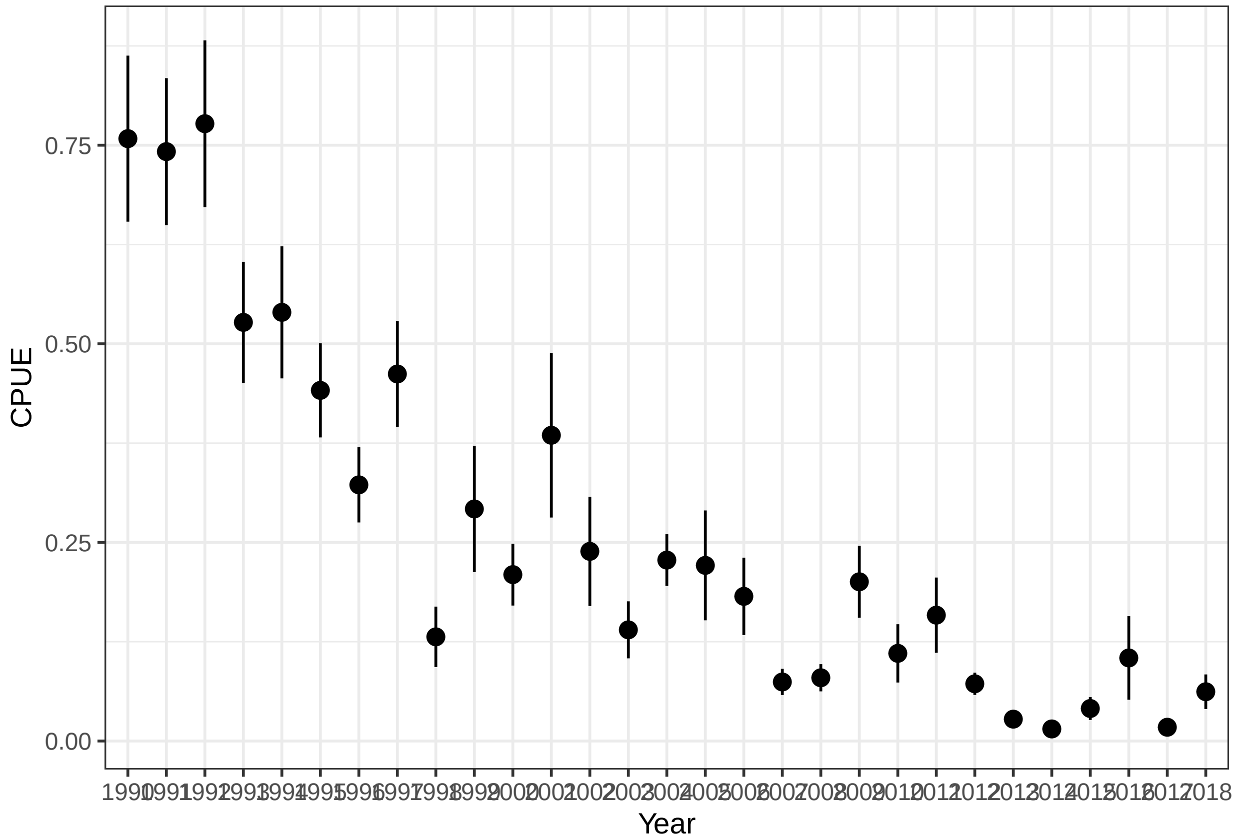
```r
bum2<-read.csv("BUM2.csv")%>%
  mutate(season=factor(season),
         Year=factor(Year),
         CPUE=BUM/sets,
         present=ifelse(BUM>0,1,0))
summary(bum2)
```

```
##        X                Year          sets            BUM
##   Min.   :   1.0    1994   : 60    Min.   : 1.00    Min.   :  0.000
```

```
##  1st Qu.: 250.8   1991    : 56   1st Qu.:14.00   1st Qu.:  0.000
##  Median : 500.5   1996    : 53   Median :20.00   Median :  2.000
##  Mean   : 500.5   1993    : 49   Mean   :19.58   Mean   :  6.075
##  3rd Qu.: 750.2   1995    : 46   3rd Qu.:20.00   3rd Qu.:  7.000
##  Max.   :1000.0   1998    : 42   Max.   :97.00   Max.   :108.000
##                   (Other):694
##       lat              lon                hbf          season       CPUE
##  Min.   :-47.50   Min.   :-53.500   Min.   : 3.00   1:252   Min.   :0.0000
##  1st Qu.:-20.50   1st Qu.: -8.750   1st Qu.:10.00   2:251   1st Qu.:0.0000
##  Median :-11.50   Median :  1.500   Median :10.00   3:214   Median :0.1000
##  Mean   :-15.53   Mean   : -1.823   Mean   :15.43   4:283   Mean   :0.3017
##  3rd Qu.: -6.50   3rd Qu.:  7.500   3rd Qu.:24.00           3rd Qu.:0.4487
##  Max.   :  0.00   Max.   : 19.500   Max.   :24.00           Max.   :4.0000
##
##     present
##  Min.   :0.000
##  1st Qu.:0.000
##  Median :1.000
##  Mean   :0.635
##  3rd Qu.:1.000
##  Max.   :1.000
##
```

```
ggplot(bum2,aes(x=Year,y=CPUE))+
  stat_summary()
```

```
## No summary function supplied, defaulting to `mean_se()`
```




```
lnorm.mean=function(x1,x1e) {
  exp(x1+0.5*x1e^2)
}

lnorm.se=function(x1,x1e) {
  ((exp(x1e^2)-1)*exp(2*x1+x1e^2))^0.5
}
lo.se=function(x1,x1e,x2,x2e) {
  if(length(x1[!is.na(x1) &!is.na(x2)])>1 & length(unique(x1))>1 & length(unique(x2))>1 )
   cor12=cor(x1[!is.na(x1) &!is.na(x2)],x2[!is.na(x1) &!is.na(x2)]) else cor12=0
  (x1e^2 * x2^2 +x2e^2*x1^2+2*x1*x2*cor12*x1e*x2e)^0.5
}
```

Part 1. binomial

```
BinomFit<-glmmTMB(present~(Year+season+hbf)^2,
                  data=bum2,
                  family="binomial",
                na.action = "na.fail")
```

```
## Warning in finalizeTMB(TMBStruc, obj, fit, h, data.tmb.old): Model convergence
## problem; non-positive-definite Hessian matrix. See vignette('troubleshooting')

## Warning in finalizeTMB(TMBStruc, obj, fit, h, data.tmb.old): Model convergence
## problem; singular convergence (7). See vignette('troubleshooting'),
## help('diagnose')
```

```r
BinDredge<-dredge(BinomFit,rank = "BIC")
```

```
## Fixed terms are "cond((Int))" and "disp((Int))"

## Warning in finalizeTMB(TMBStruc, obj, fit, h, data.tmb.old): Model convergence
## problem; singular convergence (7). See vignette('troubleshooting'),
## help('diagnose')

## Warning in finalizeTMB(TMBStruc, obj, fit, h, data.tmb.old): Model convergence
## problem; non-positive-definite Hessian matrix. See vignette('troubleshooting')

## Warning in finalizeTMB(TMBStruc, obj, fit, h, data.tmb.old): Model convergence
## problem; singular convergence (7). See vignette('troubleshooting'),
## help('diagnose')
## Warning in finalizeTMB(TMBStruc, obj, fit, h, data.tmb.old): Model convergence
## problem; singular convergence (7). See vignette('troubleshooting'),
## help('diagnose')

## Warning in finalizeTMB(TMBStruc, obj, fit, h, data.tmb.old): Model convergence
## problem; non-positive-definite Hessian matrix. See vignette('troubleshooting')

## Warning in finalizeTMB(TMBStruc, obj, fit, h, data.tmb.old): Model convergence
## problem; singular convergence (7). See vignette('troubleshooting'),
## help('diagnose')

## Warning in finalizeTMB(TMBStruc, obj, fit, h, data.tmb.old): Model convergence
## problem; non-positive-definite Hessian matrix. See vignette('troubleshooting')

## Warning in finalizeTMB(TMBStruc, obj, fit, h, data.tmb.old): Model convergence
## problem; singular convergence (7). See vignette('troubleshooting'),
## help('diagnose')
```

```r
head(BinDredge)
```

```
## Global model call: glmmTMB(formula = present ~ (Year + season + hbf)^2, data = bum2,
##     family = "binomial", na.action = "na.fail", ziformula = ~0,
##     dispformula = ~1)
## ---
## Model selection table
##    cnd((Int)) dsp((Int)) cnd(hbf) cnd(ssn) cnd(Yer) cnd(hbf:ssn) df   logLik
## 4     1.23300          +  0.03894        +                        5 -591.447
## 3     1.85800          +                 +                        4 -602.916
## 12    0.84220          +  0.06645        +                   +    8 -589.369
## 8     2.32800          +  0.03874        +        +              33 -530.231
## 2    -0.08169          +  0.04232                                 2 -641.085
## 7     2.94800          +                 +        +              32 -539.785
##       BIC delta weight
## 4  1217.4  0.00  0.999
## 3  1233.5 16.03  0.000
## 12 1234.0 16.57  0.000
```

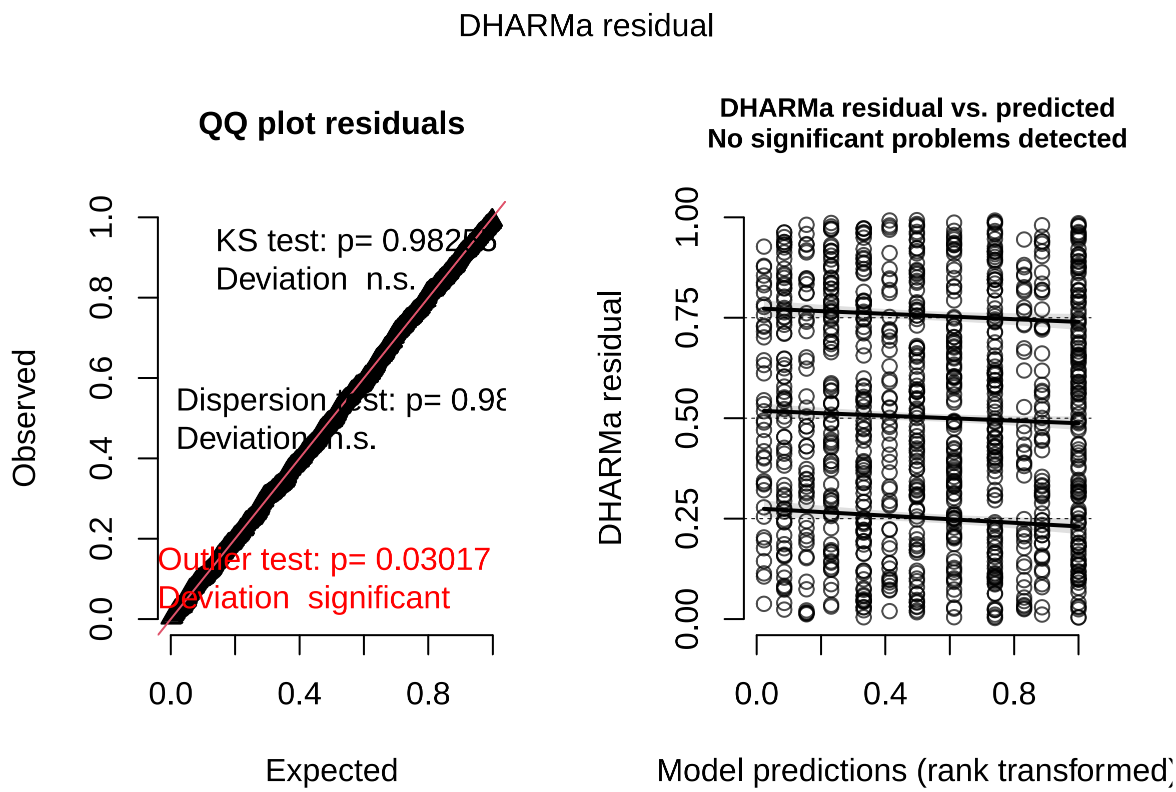
```
## 8  1288.4 70.99  0.000
## 2  1296.0 78.55  0.000
## 7  1300.6 83.18  0.000
## Models ranked by BIC(x)
```

```r
BinBest<-get.models(BinDredge,subset=1)[[1]]
summary(BinBest)
```

```
##  Family: binomial  ( logit )
## Formula:          present ~ hbf + season + 1
## Data: bum2
##
##      AIC      BIC   logLik -2*log(L)  df.resid
##   1192.9   1217.4   -591.4    1182.9       995
##
##
## Conditional model:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  1.233218   0.222566   5.541 3.01e-08 ***
## hbf          0.038943   0.008192   4.754 2.00e-06 ***
## season2     -1.547734   0.226176  -6.843 7.75e-12 ***
## season3     -2.068711   0.232364  -8.903  < 2e-16 ***
## season4     -1.210910   0.224318  -5.398 6.73e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
plot(simulateResiduals(BinBest))
```

```
## DHARMa:testOutliers with type = binomial may have inflated Type I error rates for integer-valued dis
```



Residuals look fine. Best model has hbf and season.

Part 2. log(CPUE) positive only

```
posdat<-filter(bum2,present==1)
cpueFit<-glmmTMB(log(CPUE)~(Year+season+hbf)^2,
                 data=posdat,
                 family="gaussian",
               na.action = "na.fail")
```

```
## dropping columns from rank-deficient conditional model: Year2000:season3, Year2005:season3, Year2015

## Warning in finalizeTMB(TMBStruc, obj, fit, h, data.tmb.old): Model convergence
## problem; function evaluation limit reached without convergence (9). See
## vignette('troubleshooting'), help('diagnose')
```

```
cpueDredge<-dredge(cpueFit,rank = "BIC")
```

```
## Fixed terms are "cond((Int))" and "disp((Int))"

## Warning in finalizeTMB(TMBStruc, obj, fit, h, data.tmb.old): Model convergence
## problem; function evaluation limit reached without convergence (9). See
## vignette('troubleshooting'), help('diagnose')
## Warning in finalizeTMB(TMBStruc, obj, fit, h, data.tmb.old): Model convergence
## problem; function evaluation limit reached without convergence (9). See
## vignette('troubleshooting'), help('diagnose')
```

```
## dropping columns from rank-deficient conditional model: season3:Year2000, season3:Year2005, season3:

## dropping columns from rank-deficient conditional model: season3:Year2000, season3:Year2005, season3:
## dropping columns from rank-deficient conditional model: season3:Year2000, season3:Year2005, season3:
## dropping columns from rank-deficient conditional model: season3:Year2000, season3:Year2005, season3:

## Warning in finalizeTMB(TMBStruc, obj, fit, h, data.tmb.old): Model convergence
## problem; function evaluation limit reached without convergence (9). See
## vignette('troubleshooting'), help('diagnose')

## dropping columns from rank-deficient conditional model: season3:Year2000, season3:Year2005, season3:

## Warning in finalizeTMB(TMBStruc, obj, fit, h, data.tmb.old): Model convergence
## problem; function evaluation limit reached without convergence (9). See
## vignette('troubleshooting'), help('diagnose')
```

```r
head(cpueDredge)
```

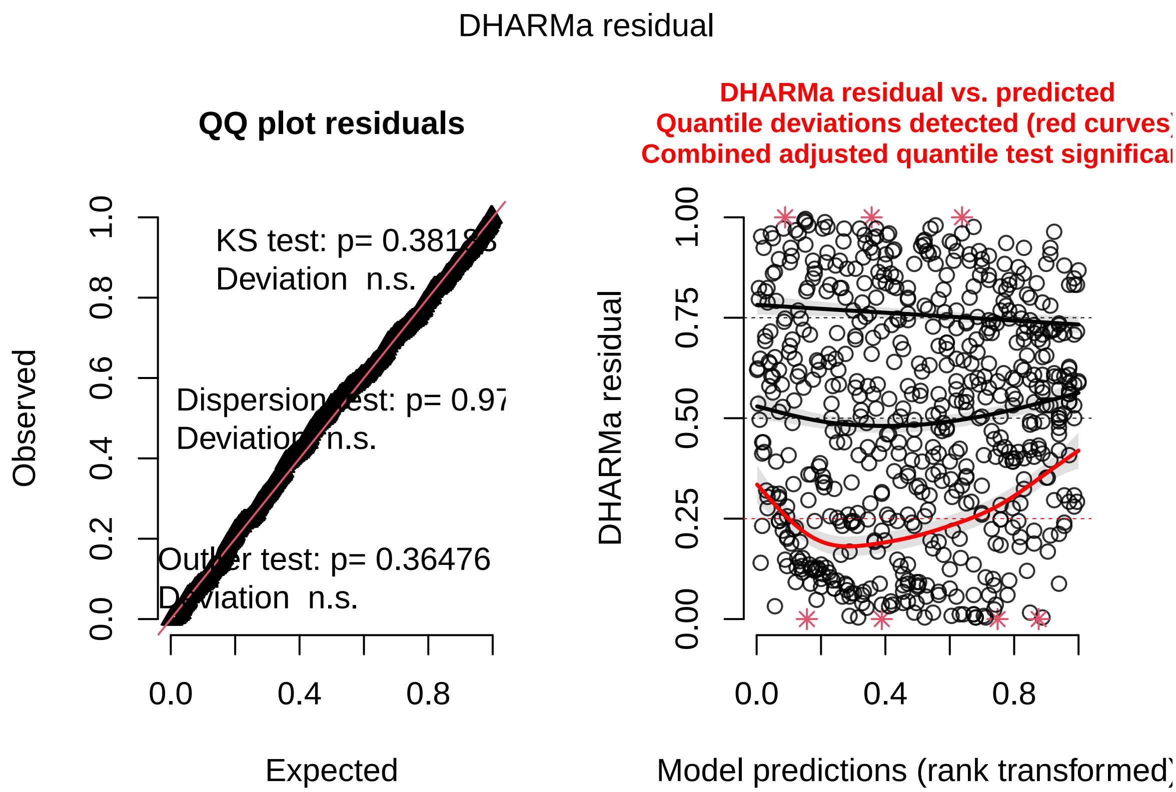
```
## Global model call: glmmTMB(formula = log(CPUE) ~ (Year + season + hbf)^2, data = posdat,
##     family = "gaussian", na.action = "na.fail", ziformula = ~0,
##     dispformula = ~1)
## ---
## Model selection table
##    cnd((Int)) dsp((Int)) cnd(hbf) cnd(ssn) cnd(Yer) cnd(hbf:ssn) df   logLik
## 8     -0.7560          +  0.02636        +        +              34 -788.841
## 16    -0.5271          +  0.01069        +        +            + 37 -780.341
## 7     -0.2745          +                 +        +              33 -807.762
## 6     -0.8921          +  0.02729                 +              31 -816.360
## 5     -0.4173          +                          +              30 -835.111
## 4     -1.6470          +  0.03564        +                        6 -919.039
##       BIC delta weight
## 8  1797.1  0.00  0.765
## 16 1799.5  2.36  0.235
## 7  1828.5 31.39  0.000
## 6  1832.8 35.68  0.000
## 5  1863.8 66.73  0.000
## 4  1876.8 79.70  0.000
## Models ranked by BIC(x)
```

```r
cpueBest<-get.models(cpueDredge,subset=1)[[1]]
summary(cpueBest)
```

```
##  Family: gaussian  ( identity )
## Formula:          log(CPUE) ~ hbf + season + Year + 1
## Data: posdat
##
##      AIC       BIC    logLik -2*log(L)  df.resid
##   1645.7    1797.1    -788.8    1577.7       601
##
##
## Dispersion estimate for gaussian family (sigma^2): 0.702
```

```
##
## Conditional model:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.756039   0.166278  -4.547 5.45e-06 ***
## hbf          0.026363   0.004222   6.245 4.25e-10 ***
## season2      0.170267   0.093473   1.822  0.06852 .
## season3     -0.481653   0.108815  -4.426 9.58e-06 ***
## season4     -0.416731   0.087323  -4.772 1.82e-06 ***
## Year1991    -0.007740   0.191127  -0.040  0.96770
## Year1992     0.065175   0.211615   0.308  0.75809
## Year1993    -0.349957   0.195932  -1.786  0.07408 .
## Year1994    -0.259636   0.190252  -1.365  0.17235
## Year1995    -0.531589   0.196994  -2.699  0.00697 **
## Year1996    -0.467193   0.201288  -2.321  0.02029 *
## Year1997    -0.495033   0.205464  -2.409  0.01598 *
## Year1998    -1.045946   0.240199  -4.354 1.33e-05 ***
## Year1999    -0.662531   0.232845  -2.845  0.00444 **
## Year2000    -1.008375   0.215595  -4.677 2.91e-06 ***
## Year2001    -1.013366   0.235493  -4.303 1.68e-05 ***
## Year2002    -1.042427   0.259686  -4.014 5.97e-05 ***
## Year2003    -1.299083   0.222097  -5.849 4.94e-09 ***
## Year2004    -1.193067   0.209245  -5.702 1.19e-08 ***
## Year2005    -0.690554   0.329878  -2.093  0.03632 *
## Year2006    -1.213068   0.235218  -5.157 2.51e-07 ***
## Year2007    -1.644254   0.237250  -6.930 4.19e-12 ***
## Year2008    -1.639884   0.236206  -6.943 3.85e-12 ***
## Year2009    -0.995445   0.248010  -4.014 5.98e-05 ***
## Year2010    -1.138099   0.272862  -4.171 3.03e-05 ***
## Year2011    -1.499316   0.239514  -6.260 3.85e-10 ***
## Year2012    -1.959548   0.230834  -8.489  < 2e-16 ***
## Year2013    -1.937776   0.328791  -5.894 3.78e-09 ***
## Year2014    -2.063724   0.371411  -5.556 2.75e-08 ***
## Year2015    -1.678323   0.347797  -4.826 1.40e-06 ***
## Year2016    -1.448164   0.314183  -4.609 4.04e-06 ***
## Year2017    -2.043245   0.402543  -5.076 3.86e-07 ***
## Year2018    -1.672846   0.314694  -5.316 1.06e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
plot(simulateResiduals(cpueBest))
```



Residuals look fine.INcludes multipel variables.

Calculate index

```
newdata3<-expand.grid(Year=factor(levels(bum2$Year)),
                    season=factor("1"),
                    hbf=mean(bum2$hbf),
                    sets=1)  #1 set to get CPUE
temp<-predict(BinBest,
              newdata = newdata3,
              se.fit = TRUE,
              type="response")
newdata3$binomial<-temp$fit
newdata3$binomial.se<-temp$se.fit
temp<-predict(cpueBest,
              newdata = newdata3,
              se.fit = TRUE) #response is on log scale
#lognormal conversion functions to get mean if positive
newdata3$pos.cpue<-lnorm.mean(temp$fit,temp$se.fit)
newdata3$pos.cpue.se<-lnorm.se(temp$fit,temp$se.fit)
#index
newdata3<-newdata3 %>%
  mutate(fit=binomial*pos.cpue,
         se=lo.se(binomial,binomial.se,pos.cpue,pos.cpue.se))
head(newdata3)
```

```
##   Year season    hbf sets  binomial binomial.se  pos.cpue pos.cpue.se       fit
```

```
## 1 1990      1 15.425    1 0.8622275  0.02206652 0.7128669  0.10594351 0.6146534
## 2 1991      1 15.425    1 0.8622275  0.02206652 0.7063183  0.09748269 0.6090071
## 3 1992      1 15.425    1 0.8622275  0.02206652 0.7626295  0.12485599 0.6575601
## 4 1993      1 15.425    1 0.8622275  0.02206652 0.5023064  0.07420887 0.4331023
## 5 1994      1 15.425    1 0.8622275  0.02206652 0.5489978  0.07546454 0.4733610
## 6 1995      1 15.425    1 0.8622275  0.02206652 0.4189263  0.06222936 0.3612098
##           se
## 1 0.09269194
## 2 0.08548511
## 3 0.10896165
## 4 0.06493789
## 5 0.06618574
## 6 0.05444637
```

Compare fits

```
ggplot(newdata3,aes(x=as.numeric(as.character(Year)),y=fit,ymin=fit-se,
           ymax=fit+se))+
  geom_point(alpha=0.8)+
  geom_errorbar(width=0.1,alpha=0.8)+
  labs(x="Year",
    y="Standardized CPUE (±SE)")
```